Unit Examination 2

Due: Wednesday, November 18 at 9:00 AM **Value: 125 points** **Format: 50-minute in-class examination**

Purpose

Integrate energy transformation, respiration, photosynthesis, DNA replication, and gene-expression evidence.

Your task

- Part A (25): Complete a five-row pathway matrix for glycolysis, citric acid cycle, oxidative phosphorylation, light reactions, and Calvin cycle. For each, give cellular location, one key input, one key product, and biological purpose.
- Part B (25): For each treatment - ATP synthase inhibitor, proton leak, complex I inhibitor, and absence of oxygen - predict oxygen consumption, proton gradient, NADH level, and ATP production. Explain one prediction in detail.
- Part C (25): Cells receive a 10-minute radioactive amino-acid pulse followed by unlabeled chase. Signal appears first in cytosol, then ER, then medium. Explain the time order, predict the result after adding a translation inhibitor at minute 0, and name one conclusion the experiment cannot support.
- Part D1 (15): Treatment Y leaves target-gene mRNA unchanged but increases protein fourfold. Give two mechanisms acting after transcription and one experiment distinguishing them.
- Part D2 (15): Treatment Z increases both mRNA and protein fourfold. Explain why this is consistent with but does not prove transcriptional activation.
- Part E (20): Design a pulse-labeling or transcription-block experiment that distinguishes reduced transcription from faster RNA degradation. Include controls and predicted time-course results for both explanations.

Submit

- Completed examination booklet and approved handwritten reference card.

Important note

Closed note except for one handwritten 4 x 6 inch reference card. Basic calculators are permitted.

Grading criteria

Criterion	Pts	Full-credit evidence
Part A	25	Correctly connects pathway steps, locations, and products.
Part B	25	Mechanistic predictions are internally consistent.
Part C	25	Accurately interprets time-course evidence.
Part D	30	Distinguishes regulatory levels with evidence.
Part E	20	Design and controls resolve the stated alternatives.